Exercise 2 · SQL fundamentals
Aggregate functions & Grouping
Database: employees.db

1 Count function
Write SQL query to find the total number of employees in the company.

```
SELECT COUNT(id) AS number_of_employees
FROM employees
```

| number_of employees |
|---|
| 10 |

2. SUM() function
Write a SQL function to find total salary paid to all employees in the IT department

```
SELECT sum(salary) AS total salary
FROM employees
WHERE department='IT';
```

| Total Salary |
|---|
| 220 000 |

3. AVG() function
Write a SQL query to calculate the average salary of employees in the HR department.

```
SELECT AVG(salary) AS average salary
FROM employees
```

where department = 'HR';

| average salary |
|---|
| 49 500 |

4. MIN() and MAX() functions

Write a SQL query to find the highest and lowest salary in the company.

```
SELECT MAX(salary) AS highest salary
from employees;
```

| highest salary |
|---|
| 62 000 |

```
SELECT MIN(salary) AS lowest salary
from employees;
```

| lowest salary |
|---|
| 45 000 |

5. GROUP BY Statement

Write a SQL query to group ~~by~~ employees by department and display the total salary paid in each department.

```
SELECT department,
       SUM(salary) AS total salary
from employees
GROUP BY department;
```

| department | total salary |
|---|---|
| IT | 220 000 |
| HR | 99 000 |
| finance | 119 000 |
| Marketing | 105 000 |

6. GROUP BY and COUNT

Write a SQL query to count how many employees work in each city

```
SELECT id, COUNT(*) AS total employees
FROM employees
GROUP BY city;
```

| City | Employees |
|---|---|
| New York | 2 |
| Chicago | 3 |
| Los Angeles | 2 |
| San Francisco | 2 |
| Houston | 1 |

7. GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate average salary in each department and order by the results in descending order of average salary.

```
SELECT department, AVG(salary) AS average.salary
FROM employees
GROUP BY department
ORDER BY average.salary DESC;
```

department average.salary
New York
Chicago
Los Angeles
San francisco
Houston

| department | average.salary |
|---|---|
| Houston | 59 000 |
| Los Angeles | 55 500 |
| Chicago | 54 333 |
| New York | 54 000 |
| San francisco | 51 000 |

8. HAVING clause

Write a SQL query to find departments where the total salary paid exceeds 100,000

(Use GROUP BY and HAVING)

```
SELECT department, sum (salary) AS total. salary
FROM employees
GROUP BY department
HAVING sum (salary) > 100 000;
```

| department | total. salary |
|---|---|
| IT | 220 000 |
| finance | 119 000 |
| Marketing | 105 000 |

9. Combining GROUP BY, HAVING and ORDER BY
Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
SELECT city, COUNT(*) AS total_employees
FROM employees
GROUP BY department
HAVING COUNT * > 1
ORDER BY total employees DESC;
```

| department | total employees |
|---|---|
| IT | 4 |
| HR | 2 |
| finance | 2 |
| Marketing | 2 |

10 Combining Aggregate functions
Write SQL query to find the department with the highest average salary.

```
SELECT department,
       AVG(salary) AS average.salary
FROM employees
GROUP BY department
ORDER BY average.salary DESC
LIMIT 1;
```

| department | average.salary |
| --- | --- |
| Houston | 59 000 |